

CSD 420 Advanced Java Programming
Module 4 Programming Assignment - LinkedList Performance
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```
(base) fernandocontreras@Fernandos-MacBook-Pro csd-420 % cd module-4

(base) fernandocontreras@Fernandos-MacBook-Pro module-4 % ls
LinkedListPerformanceTest.java

(base) fernandocontreras@Fernandos-MacBook-Pro module-4 % javac LinkedListPerformanceTest.java

(base) fernandocontreras@Fernandos-MacBook-Pro module-4 % java LinkedListPerformanceTest
=== Testing with 50,000 integers ===
List size: 50,000
Iterator traversal: 1 ms (checksum 1249975000)
get(index) traversal: 869 ms (checksum 1249975000)
Test PASSED: both traversals produced the same sum

=== Testing with 500,000 integers ===
List size: 500,000
Iterator traversal: 3 ms (checksum 124999750000)
get(index) traversal: 90941 ms (checksum 124999750000)
Test PASSED: both traversals produced the same sum
```

Results Analysis

The observed timings from the two test runs are summarized below:

List size	Iterator time	get(index) time	get(index) slower by
50,000	1 ms	869 ms	~870x
500,000	3 ms	90,941 ms (~1.5 min)	~30,300x

The difference between the two approaches is dramatic and grows quickly as the list gets larger. Multiplying the list size by ten made the Iterator traversal only about three times slower, which is roughly linear scaling as expected for an $O(n)$ operation. In contrast, the `get(index)` traversal became about 104 times slower for the same ten times increase in list size, which is almost exactly the quadratic scaling expected for an $O(n^2)$ operation ($10^2 = 100$).